What is Regression?

Predicting a continuous number from input features

Regression is a supervised learning technique where the goal is to predict a continuous numerical value (output) from one or more input features. Unlike classification (which predicts a category), regression predicts a quantity — a house price, tomorrow's temperature, or an employee's expected salary

Input Features (X)	Output to Predict (y)	Real Example
House area, location	Sale price ($)	Zillow / Magicbricks
Years of experience	Annual salary	LinkedIn salary tool
Date, humidity, wind	Temperature (°C)	Weather forecast
Rainfall, soil type	Crop yield (tonnes/acre)	AgriTech models
Ad spend	Revenue generated	Marketing analytics

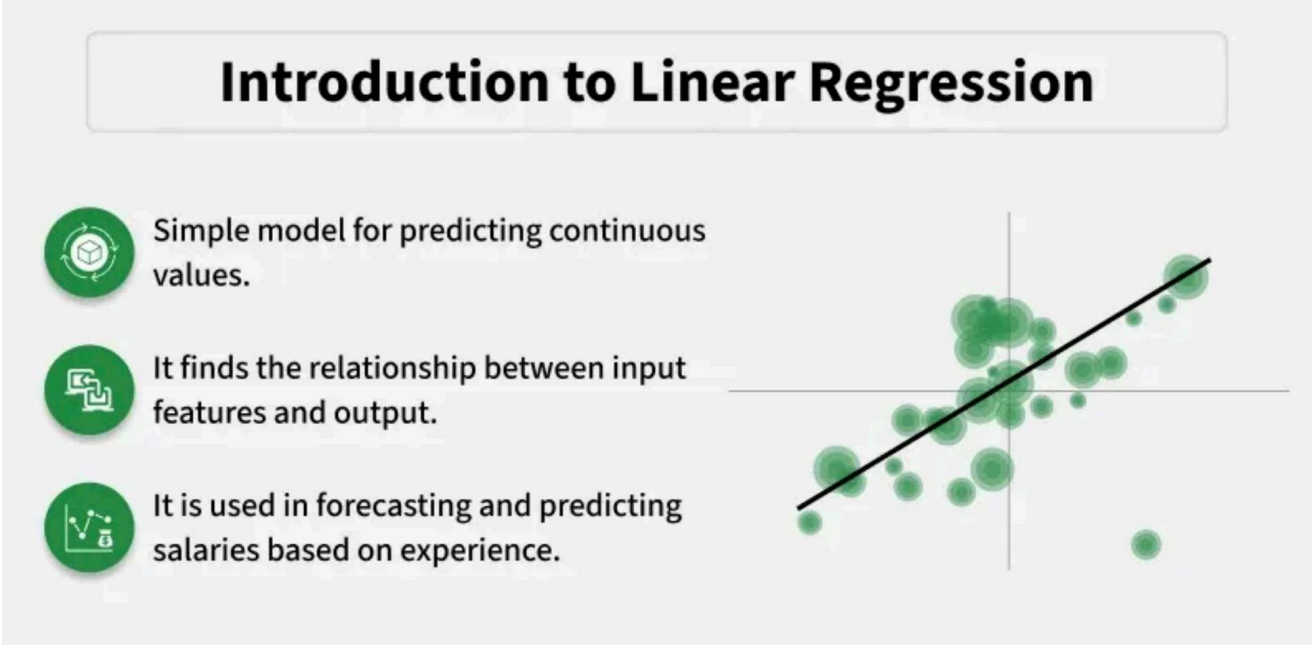
The main goal of regression is:

Understand the relationship between input and output Predict future values using that relationship

The fitted line tries to stay as close as possible to all the data points together, not just one point. Once the model learns the pattern, we can give it a new input value and it will estimate the corresponding outpu

Regression is a machine learning technique used to predict continuous numerical values based on patterns found in existing data.

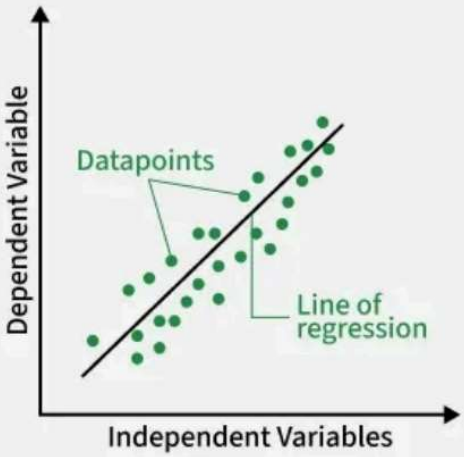
Linear regression is a type of supervised machine-learning algorithm that learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets. It assumes that there is a linear relationship between the input and output, meaning the output changes at a constant rate as the input changes. This relationship is represented by a straight line.



How Does Linear Regression Work?

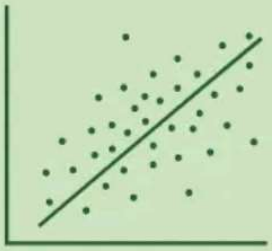
Finds the best-fitting line by minimizing prediction errors (least squares method)

It calculates coefficients for variables that minimize the error in predictions.



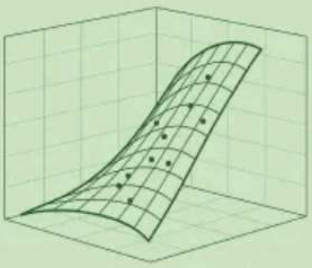
Types of Linear Regression

Simple Linear Regression



Predicts the dependent variable using a single independent variable.

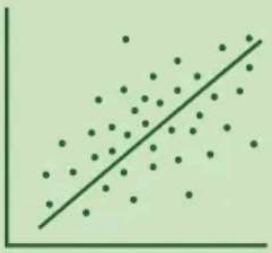
Multiple Linear Regression



Uses two or more independent variables to predict the dependent variable.

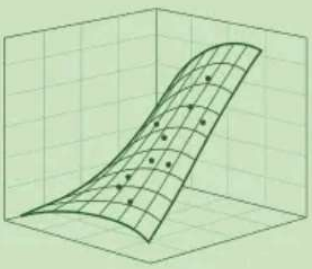
Types of Linear Regression

Simple Linear Regression



Predicts the dependent variable using a single independent variable.

Multiple Linear Regression



Uses two or more independent variables to predict the dependent variable.

For example we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up.

In the example of predicting exam scores based on hours studied.Here

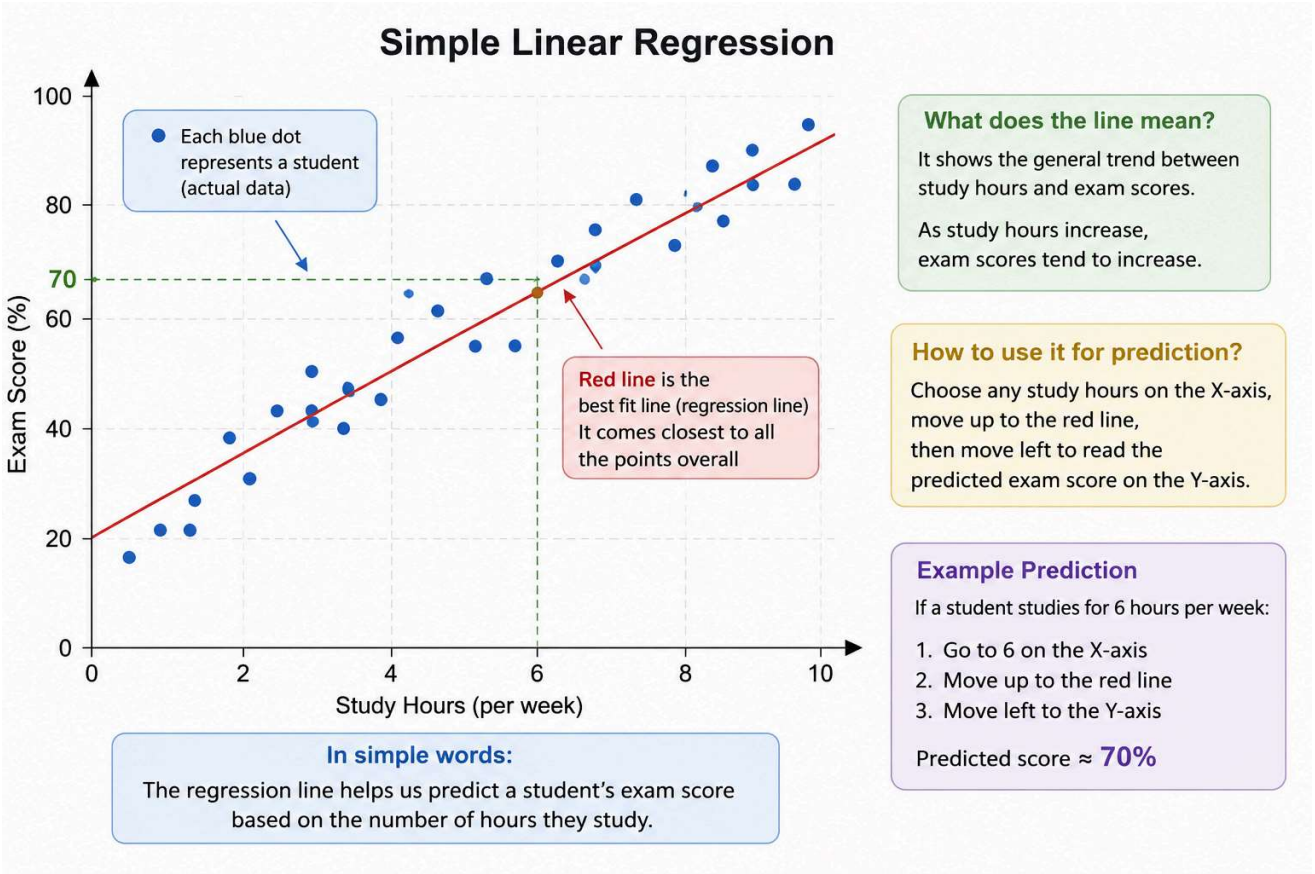
Independent variable(input): Hours studied because it's the factor we control or observe.

Dependent variable (output): Exam score because it depends on hobw many hours were studied.

We use the independent variable to predict the dependent variable.

THE BEST FIT LINE IN LINEAR REGRESSION

In linear regression, the best-fit line is the straight line that most accurately represents the relationship between the independent variable (input) and the dependent variable (output). It is the line that minimizes the difference between the actual data points and the predicted values from the model.



Imagine plotting study hours on the X-axis and exam scores on the Y-axis. Each student is a dot. A regression model draws the single straight line that comes closest to all the dots simultaneously. That line lets us predict any student's score just by knowing their study hours.

y = m·x + c

y = output | x = input | m = slope | c = intercept

```
import numpy as np

# Input data
hours = np.array([1,2,3,4,5,6,7,8,9,10])
scores = np.array([60,70,80,90,100,110,120,130,140,150])

# Linear Regression tries to find the BEST FIT LINE
# Equation of line:
# score = m * hours + c
#
# m = slope of the line
# c = intercept of the line

# Manual understanding of slope:
# slope = change in y / change in x
# slope ≈ (150 - 60) / (10 - 1)
# slope ≈ 90 / 9
```

```
# slope ≈ 10

# Intercept:
# c = y - mx
# c = 60 - (10 * 1)
# c = 50

# Final best fit equation:
# predicted_score = 10 * hours + 50

# Predict score for 5 study hours
predicted_score = 10 * 5 + 50

print("Best Fit Line Equation:")
print("score = 10 * hours + 50")

print("\nFor 5 hours → predicted score:", predicted_score)
```

```
For 5 hours → predicted score: 68.4
```

Each value in hours represents study hours.

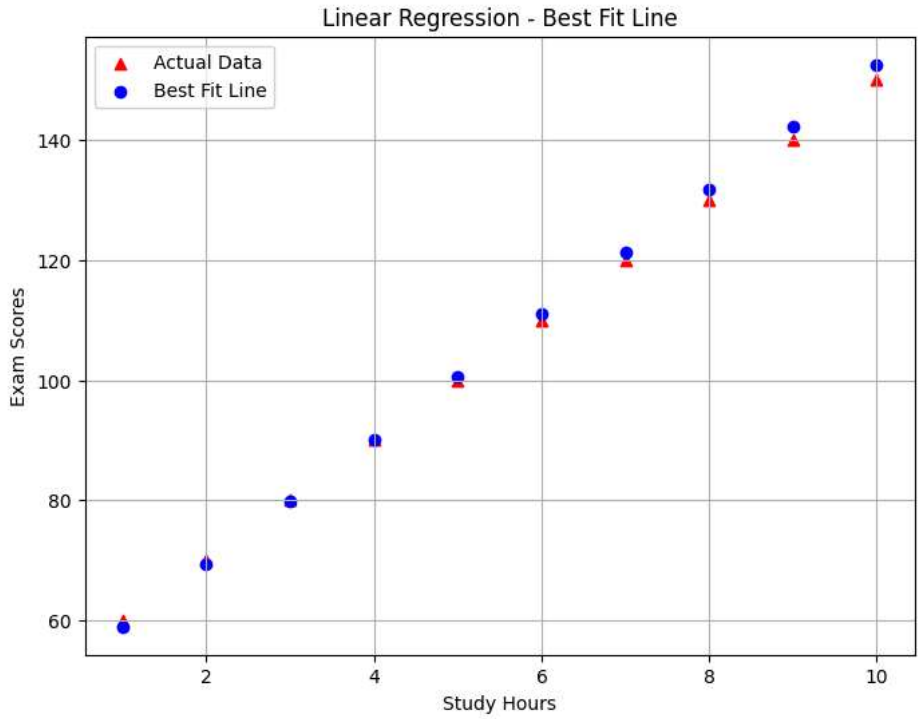
Each value in scores represents exam marks.

Linear Regression finds the best fit line that represents the relationship between these two variables.

The slope (m) tells how much the score increases when study hours increase by 1.

The intercept (c) is the starting point of the line.

```
import numpy as np
import matplotlib.pyplot as plt

# Original data
hours = np.array([1,2,3,4,5,6,7,8,9,10])
scores = np.array([60,70,80,90,100,110,120,130,140,150])

# Best Fit Line Equation
# y = 10x + 50

predicted_scores =10.41* hours + 48.52

# Create graph
plt.figure(figsize=(8,6))

# Actual data points
plt.scatter(hours, scores, label="Actual Data" ,marker="^",color="red")

# Best fit regression line
plt.scatter(hours, predicted_scores, label="Best Fit Line",color="blue")

# Labels and title
plt.xlabel("Study Hours")
plt.ylabel("Exam Scores")
plt.title("Linear Regression - Best Fit Line")

# Show legend
plt.legend()

# Show graph
plt.grid(True)
plt.show()
```



HOW THE MODEL LEARNS

A regression model learns by adjusting its slope (m) and intercept (c) so that predictions become as accurate as possible. This happens by minimising a loss function — a number that measures how wrong the model currently is.

ERROR/PREDICTION

Error = Predicted − Actual

Positive → over-predicted | Negative → under-predicted

By above calculations we will get equation as

y=mx+c

m= slope

x=hours

c= interception

Suppose:

Study Hours = 5 Actual Score = 100

If the model currently uses:

m = 8

c = 40

Then prediction becomes:

y=8(5)+40=80

So:

Actual Score = 100

Predicted Score = 80

The model is wrong by:

Error=Actual−Predicted=100−80=20

This difference is called Error.

Mean Squared Error (MSE)

$$MSE = (1/n) \times \Sigma (y\_pred - y\_actual)^2$$

n = number of samples | Σ = sum over all samples

The model checks errors for all data points.

Instead of simply adding errors, we square them:

Negative values become positive Large mistakes get punished more

y_actual ⟶ real value

y_predicted ⟶ model prediction

n ⟶ number of data points

for checking mse values we need to run above example for multiple times to check whether it is working properly or not

if we take new data as

Step-1: find predicted values

hours[5,8,7]

Score[100 , 120, 90]

Student 1

Study Hours = 5

Prediction:

y=8(5)+40=80

So:

Actual = 100

Predicted = 80

calculate for remaining students

Step 2 — Calculate Error

Error=Actual−Predicted

Student 1

100−80=20

Ccalculate for remaining students

Step 3 — Square the Error

Formula:

Error 2 =(Actual−Predicted) 2

Student 1

20 2 =400

calculate for remaining students

Actual	Predicted	Error	Error²
100	80	20	400
120	104	16	256
90	96	-6	36

GRADIENT DESCENT

Gradient Descent is the algorithm that actually updates m and c to reduce MSE. Imagine you're blindfolded on a hilly landscape and want to reach the lowest point (minimum MSE). You feel the slope under your feet and take a small step downhill. Repeat until you can't go lower.

Gradient Descent works like this:

1.Start with random values of m and c

2.Calculate predictions

3.Compute error and MSE

4.Adjust m and c

5.Repeat again and again until error

6.becomes very small

Imagine you are trying to find the coldest spot in a dark room.

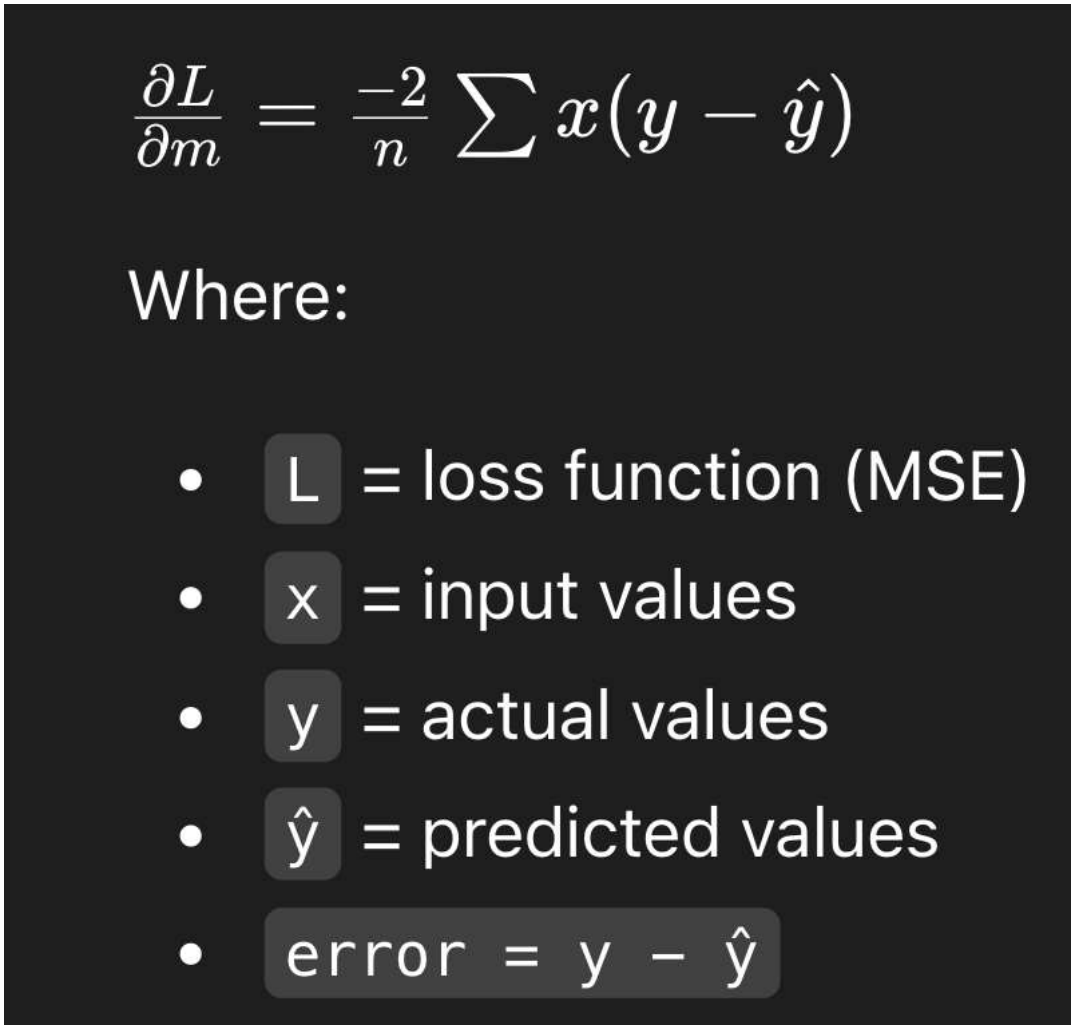
You cannot see anything, but you can feel the temperature.

If you move in one direction and it becomes colder, you continue moving that way. If it becomes warmer, you change direction.

You keep taking small steps again and again until you reach the coldest place in the room.

Real Life	Gradient Descent
Dark room	Machine learning problem
Coldest spot	Minimum error
Feeling temperature	Calculating loss/error
Small movements	Updating parameters
Reaching coldest area	Finding best fit model

Formula



LETS TRY WITH EXAMPLE CONSIDER ALL VALUES THAT ARE CALCULATED AS ZERO .

```
import numpy as np

# ----------------------------
# Dataset
# ----------------------------
```

```
# x = study hours
# y = actual scores

x = np.array([1,2,3,4,5], dtype=float)
y = np.array([60,70,80,90,100], dtype=float)

# ----------------------------
# Initial values of m and c
# The model initially knows nothing.
# ----------------------------
m = 0
c = 0

# Learning rate
learning_rate = 0.01

# Number of iterations
epochs = 1000

# Total number of data points
n = len(x)

# ----------------------------
# Gradient Descent
# ----------------------------

for i in range(epochs):

    # Predicted values
    y_pred = m * x + c

    # Error
    error = y - y_pred

    # Mean Squared Error (MSE)
    mse = np.mean(error ** 2)

    # Gradients
    dm = (-2/n) * np.sum(x * error)#How much the slope (m) should change
    dc = (-2/n) * np.sum(error)#How much the intercept (c) should change

    # Update m and c
    m = m - learning_rate * dm
    c = c - learning_rate * dc

    # Print progress every 100 iterations
    if i % 100 == 0:
        print(f"Epoch {i}")
        print(f"MSE = {mse:.2f}")
        print(f"m = {m:.2f}, c = {c:.2f}")
        print("----------------------")

# Final equation
print("\nFinal Best Fit Line:")
print(f"y = {m:.2f}x + {c:.2f}")
```

```
Epoch 0
MSE = 6600.00
m = 5.20, c = 1.60
----------------------
Epoch 100
MSE = 177.90
m = 18.63, c = 18.84
----------------------
Epoch 200
MSE = 90.37
m = 16.15, c = 27.79
----------------------
Epoch 300
MSE = 45.90
m = 14.38, c = 34.17
----------------------
Epoch 400
MSE = 23.32
m = 13.12, c = 38.72
----------------------
Epoch 500
MSE = 11.84
m = 12.23, c = 41.96
----------------------
Epoch 600
MSE = 6.02
m = 11.59, c = 44.27
----------------------
Epoch 700
```

```
MSE = 3.06
m = 11.13, c = 45.92
----------------------
Epoch 800
MSE = 1.55
m = 10.81, c = 47.09
----------------------
Epoch 900
MSE = 0.79
m = 10.57, c = 47.93
----------------------

Final Best Fit Line:
y = 10.41x + 48.52
```

Building Models with sklearn

Train / Test Split — Why it matters If you train and test on the same data, the model's score is dishonest — it has memorised the answers.

We always keep a portion of data unseen during training so the final score reflects real-world performance

■ Golden Rule

80% of data → Training set (model learns from this). 20% of data → Test set (we evaluate on this).

Never let test data touch the training phase!

```
from sklearn.model_selection import train_test_split
import numpy as np
```

```
# Example: salary prediction from experience
experience = np.array([1,2,3,4,5,6,7,8,9,10,11,12,13,14,15])
salary = np.array([35,38,42,48,55,60,65,70,76,82,87,92,96,100,105])
```

```
X = experience.reshape(-1, 1) # sklearn expects 2D array for features
y = salary
```

```
X_train, X_test, y_train, y_test = train_test_split(
 X, y,
 test_size=0.2, # 20% for testing
 random_state=42 # ensures same split every run
)
```

```
print(f"Training samples : {len(X_train)}") # 12
print(f"Testing samples : {len(X_test)}") # 3
```

```
Training samples : 12
Testing samples : 3
```

Fitting and Predicting

```
from sklearn.linear_model import LinearRegression
# 1. Create the model object
model = LinearRegression()
```

```
# 2. Train (fit) on training data
model.fit(X_train, y_train)

# 3. Inspect what the model learned
print(f"Slope (coefficient) : {model.coef_[0]:.2f}")#slope=coef
print(f"Intercept : {model.intercept_:.2f}")

# Output: Slope ≈ 4.86 → every extra year ≈ +$4 860 salary
# 4. Predict on TEST data (unseen!)
y_pred = model.predict(X_test)

# 5. See predictions vs actual
for actual, predicted in zip(y_test, y_pred):
 print(f"Actual: {actual:>5} | Predicted: {predicted:>7.1f} | "
 f"Error: {predicted - actual:+.1f}")
```

```
Slope (coefficient) : 5.22
Intercept : 27.94
Actual:    82 | Predicted:    80.2 | Error: -1.8
Actual:    92 | Predicted:    90.6 | Error: -1.4
Actual:    35 | Predicted:    33.2 | Error: -1.8
```

Evaluation Metrics

After predicting, we need to **measure quality** using three main metrics:

Metric	Formula	Meaning	Ideal
R² Score	1 – SS_res/SS_tot	% of variance explained (0–1)	Close to 1.0
MAE	mean(|y_pred – y_actual|)	Average absolute error	Close to 0
RMSE	sqrt(mean((y_pred – y_actual)²))	Like MAE but penalises big errors	Close to 0

Metric	What it tells	Best for
R² Score	How well model explains data	Model performance
MAE	Average error	Simple interpretation
RMSE	Penalizes large errors	Detecting big mistakes

calculating all three matrics

```
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import numpy as np
y_actual = np.array([500, 300, 700, 450, 600])
y_pred = np.array([480, 320, 690, 460, 590])
#How much of the variation in data is explained by the model
r2 = r2_score(y_actual, y_pred)
#Average of absolute errors (ignores + / - signs)
mae = mean_absolute_error(y_actual, y_pred)
#RMSE (Root Mean Squared Error)
#Average error but gives more weight to big mistakes
rmse = np.sqrt(mean_squared_error(y_actual, y_pred))
print(f"R² Score : {r2:.4f} → model explains {r2*100:.1f}% of variance")
print(f"MAE : {mae:.1f} → avg error ± {mae:.0f} units")
print(f"RMSE : {rmse:.1f} → penalised avg error {rmse:.0f} units")
# Interpretation:
# R² = 0.95 means 95% of the price variation is explained by our features
# MAE = 15000 means we're off by ~$15,000 on average
# RMSE > MAE means there are some larger individual errors
```

Metric	What it tells	Best use
R² Score	How good the model is overall	Model quality
MAE	Average error	Easy interpretation
RMSE	Penalizes big errors	When large mistakes matter

Multiple Features

– Predicting Salary Real problems have multiple input features. Linear regression handles this naturally: it learns one weight (coefficient) per feature

What are Multiple Features?

in real life, salary depends on many things:

Experience

Education level

Skills

City

Company

Certifications

salary = w1·experience + w2·education + w3·age + b

Each feature gets its own weight (importance)

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error
import numpy as np
# Create a realistic salary dataset
np.random.seed(42)
n = 100
data = pd.DataFrame({
 'experience': np.random.randint(0, 20, n), # 0–20 years
 'education': np.random.randint(12, 20, n), # 12–20 years schooling
 'age': np.random.randint(22, 55, n), # 22–55 years
})
# Generate salary with some noise
data['salary'] = (
 data['experience'] * 4500 +
 data['education'] * 2000 +
 data['age'] * 300 +
 np.random.normal(0, 3000, n) +
 10000
)
# Prepare features and target
X = data[['experience', 'education', 'age']]
y = data['salary']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
# Print learned weights
for feat, coef in zip(X.columns, model.coef_):
 print(f" {feat:12s}: ${coef:,.0f} per unit increase")
print(f" Base salary (intercept): ${model.intercept_:,.0f}")
y_pred = model.predict(X_test)
print(f"\nR² : {r2_score(y_test, y_pred):.3f}")
print(f"MAE : ${mean_absolute_error(y_test, y_pred):,.0f}")
```

```
 experience  : $4,534 per unit increase
 education   : $2,203 per unit increase
 age         : $266 per unit increase
 Base salary (intercept): $7,694

R² : 0.988
MAE : $2,219
```

LAB

Step-1: create the data set

Generate Realistic House Price Data

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
np.random.seed(42)
n = 200
locations = ['Urban', 'Suburban', 'Rural']
df = pd.DataFrame({
```

```
 'area_sqft': np.random.randint(500, 4000, n),
 'num_rooms': np.random.randint(1, 7, n),
 'location': np.random.choice(locations, n),
 'age_years': np.random.randint(0, 40, n),
})
# Location multipliers
loc_map = {'Urban': 1.5, 'Suburban': 1.0, 'Rural': 0.7}#expensiv medium cheap
noise = np.random.normal(0, 20000, n)
df['price'] = (
 df['area_sqft'] * 150 +
 df['num_rooms'] * 25000 -
 df['age_years'] * 2000 +
 df['location'].map(loc_map) * 50000 +
 noise
).astype(int)
print(df.head(8).to_string(index=False))
print(f"\nDataset shape : {df.shape}")
print(df.describe()[['area_sqft','num_rooms','age_years','price']].round(0))
```

```
 area_sqft  num_rooms location  age_years  price
      3674          4    Rural         20 646067
      1360          4    Urban         35 300479
      1794          6    Urban          9 501609
      1630          6    Rural         36 355668
      1595          3    Urban          8 391991
      3592          2    Rural         23 566413
      2138          4 Suburban         34 410974
      2669          1    Urban         34 459191

Dataset shape : (200, 5)
       area_sqft  num_rooms  age_years     price
count      200.0      200.0      200.0     200.0
mean      2334.0        3.0       20.0  451548.0
std        992.0        2.0       11.0  154871.0
min        521.0        1.0        0.0   86373.0
25%       1519.0        2.0       10.0  326188.0
50%       2372.0        3.0       20.0  458282.0
75%       3230.0        5.0       30.0  575910.0
max       3999.0        6.0       39.0  781261.0
```

Step 2 — Encode & Prepare Features

```
# Encode location: Urban=2, Suburban=1, Rural=0
#Creates a tool that converts text → numbers
le = LabelEncoder()
#transform Converts them into numbers automatically
df['location_enc'] = le.fit_transform(df['location'])
# Select features and target
features = ['area_sqft', 'num_rooms', 'age_years', 'location_enc']
X = df[features]
y = df['price']
# Peek at encoded data
print(df[['location', 'location_enc']].drop_duplicates().sort_values('location_enc'))
print(f"\nFeature matrix shape : {X.shape}")
print(f"Target vector shape : {y.shape}")
```

```
   location  location_enc
0     Rural             0
6  Suburban             1
1     Urban             2

Feature matrix shape : (200, 4)
Target vector shape : (200,)
```

Step 3 — Train / Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
 X, y, test_size=0.2, random_state=42
)
print(f"Training samples : {len(X_train)} ({len(X_train)/len(X)*100:.0f}%)")
print(f"Testing samples : {len(X_test)} ({len(X_test)/len(X)*100:.0f}%)")
print(f"Feature names : {features}")
```

```
Training samples : 160 (80%)
Testing samples : 40 (20%)
Feature names : ['area_sqft', 'num_rooms', 'age_years', 'location_enc']
```

Step 4 — Train the Model

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
print("Learned Coefficients:")
print("-" * 45)
for feat, coef in zip(features, model.coef_):
 direction = "↑" if coef > 0 else "↓"
 print(f" {feat:15s}: Rs {coef:>10,.0f} {direction}")
print(f" {'intercept':15s}: Rs {model.intercept_:>10,.0f}")
print("-" * 45)
# Interpretation:
# area_sqft → every extra sq ft adds ~Rs 150 to price
# num_rooms → every extra room adds ~Rs 25,000
# age_years → every year older reduces by ~Rs 2,000
```

```
Learned Coefficients:
---------------------------------------------
 area_sqft      : Rs        151 ↑
 num_rooms      : Rs     23,169 ↑
 age_years      : Rs     -2,262 ↓
 location_enc   : Rs     18,114 ↑
 intercept      : Rs     46,789
---------------------------------------------
```

Step 5 — Evaluate

```
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
print("=" * 40)
print(" MODEL EVALUATION REPORT")
print("=" * 40)
print(f"R² Score : {r2:.4f} {'✓ Excellent' if r2>0.9 else '✓ Good' if r2>0.8 else '■ Needs work'}")
print(f"MAE : Rs {mae:>10,.0f}")
print(f"RMSE : Rs {rmse:>10,.0f}")
print("=" * 40)
# Show first 8 predictions vs actuals
print("\nSample Predictions:")
print(f"{'Actual':>10} | {'Predicted':>10} | {'Error':>10}")
print("-" * 36)
for a, p in zip(y_test[:8], y_pred[:8]):
 print(f"{a:>10,} | {p:>10,.0f} | {p-a:>+10,.0f}")
```

```
========================================
 MODEL EVALUATION REPORT
========================================
R² Score : 0.9872 ✓ Excellent
MAE : Rs     13,439
RMSE : Rs     16,959
========================================

Sample Predictions:
    Actual |  Predicted |      Error
------------------------------------
   489,899 |    478,320 |    -11,579
   601,329 |    572,877 |    -28,452
   623,788 |    650,489 |    +26,701
   291,458 |    271,086 |    -20,372
   390,040 |    344,431 |    -45,609
   123,884 |    134,545 |    +10,661
   484,442 |    443,295 |    -41,147
   562,823 |    556,350 |     -6,473
```

Step 6 — Predict on New Houses

```
# New houses to price
new_houses = pd.DataFrame({
 'area_sqft': [1200, 2500, 800, 3500],
 'num_rooms': [3, 5, 1, 6 ],
 'age_years': [5, 10, 25, 2 ],
 'location_enc':[2, 1, 0, 2 ], # 2=Urban,1=Suburban,0=Rural
 'location_lbl':['Urban','Suburban','Rural','Urban'],
})
predictions = model.predict(new_houses[features])
print("\n HOUSE PRICE PREDICTIONS")
print("=" * 60)
for i, (_, row) in enumerate(new_houses.iterrows()):
 print(f" House {i+1}: {row.area_sqft} sqft | "
 f"{row.num_rooms} rooms | "
 f"{row.age_years}yr old | "
 f"{row.location_lbl}")
 print(f" Estimated Price → Rs {predictions[i]:>10,.0f}")
 print()
```

```
HOUSE PRICE PREDICTIONS
============================================================
House 1: 1200 sqft | 3 rooms | 5yr old | Urban
Estimated Price → Rs    321,914

House 2: 2500 sqft | 5 rooms | 10yr old | Suburban
Estimated Price → Rs    534,588

House 3: 800 sqft | 1 rooms | 25yr old | Rural
Estimated Price → Rs    133,872

House 4: 3500 sqft | 6 rooms | 2yr old | Urban
Estimated Price → Rs    744,552
```

Cheat sheat

DATA PREPARATION

Task	Code	Purpose
Import NumPy	`import numpy as np`	Numerical operations
Import Pandas	`import pandas as pd`	Handle dataset (tables)
Create DataFrame	`df = pd.DataFrame({...})`	Build dataset

Feature & Target Selection Task

Task	Code	Purpose
Features (X)	`X = df[['col1','col2']]`	Input variables
Target (y)	`y = df['target']`	Output variable

Train-Test Split

Task	Code	Purpose
Import split	`from sklearn.model_selection import train_test_split`	Split data
Split data	`train_test_split(X,y,test_size=0.2,random_state=42)`	Train & test

Model Creation & Training

Task	Code	Purpose
Import model	`from sklearn.linear_model import LinearRegression`	Load algorithm
Create model	`model = LinearRegression()`	Initialize model
Train model	`model.fit(X_train, y_train)`	Learn patterns

Prediction

Task	Code	Purpose
Predict	`y_pred = model.predict(X_test)`	Predict unseen data

Model Evaluation

Metric	Code	Meaning
R^2 Score	`r2_score(y_test, y_pred)`	How well model explains data
MAE	`mean_absolute_error(y_test, y_pred)`	Average error
RMSE	`np.sqrt(mean_squared_error(y_test, y_pred))`	Penalizes large errors

Model Parameters

Task	Code	Meaning
Slope (weights)	`model.coef_`	Importance of each feature
Intercept	`model.intercept_`	Base value

Data Encoding (Categorical → Numeric)

Task	Code	Purpose
Label Encoding	`LabelEncoder()`	Convert text to numbers
Fit transform	`le.fit_transform(df['col'])`	Encode categories

Data Reshaping

Task	Code	Meaning
Reshape column	`reshape(-1,1)`	Convert 1D → 2D
Auto rows	`-1`	Let NumPy decide rows
Single feature	`1`	One column

Noise

Concept	Code	Meaning
Noise	`np.random.normal(0, 3000, n)`	Random variation